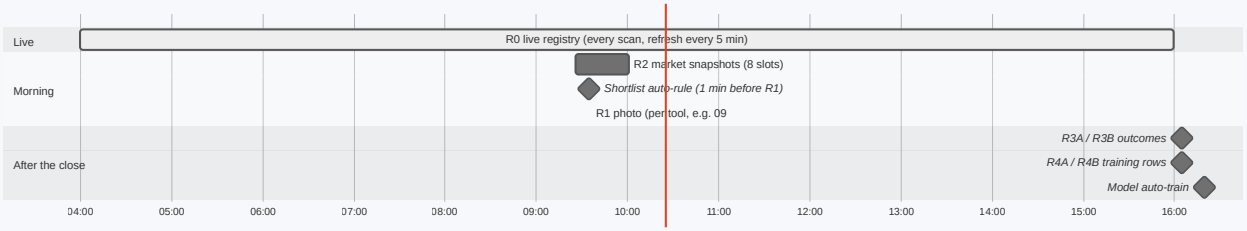
The registers

What each tool writes down, when, where it is stored, and every column. A register is how a day of screening becomes data that qp can backtest, the scorecard can grade, and the model can learn from. Written from `src/warehouse/registers.js`, `src/sideH/capture.js`, `src/scheduler.js` and `src/db/index.js`.

How the numbers get onto a row in the first place is in DATA-PIPELINE.md. Each tool has its own database (`data/tradedesk.db` for T1, `data/t<N>.db` for the others), so every register below exists once per tool.

The day, in registers



Register	Written	Trigger	Table	One row per
R0	continuously	every scan and refresh	memory + `r0_checkpoint`	ticker, today
R1	once, at the tool's capture minute	scheduler `R1 Capture HH:MM`	`r1_frozen`	ticker, date
R2	09:26, 09:31, 09:36, 09:41, 09:46, 09:51, 09:56, 10:01	scheduler	`r2_market_snapshots`	slot, date
R3A / R3B	16:05	scheduler `R3 EOD Capture`	`r3a` / `r3b`	ticker, date
R4A / R4B	read time (R1 joined to R3) and 16:05 into the training tables	R3 capture	`r4a_train` / `r4b_train`	ticker, date
Shortlist	on every star and at the auto-rule minute	the card's ☆, the auto-rule	`shortlist`	date (a list of items)

Capture times per tool (`tools.config.json`, editable on Manage tools):

Tool	R1 photo	Entry A	Entry B
T1, T7 (and T3, T9)	09:36	09:37	09:40
T2, T11 (and T5, T8, T10)	09:46	09:47	09:51
T4	10:01	10:02	10:06
T6	10:16	10:17	10:21 (see D-S9: before T6's 11:30 window)

R0 — the live registry

What the screen shows. Built by every scan (DATA-PIPELINE.md), kept in the process's memory, checkpointed to the database after each scan, reloaded on a same-day restart, cleared at midnight.

- **Holds every card of today:** live ones (matched by a screener this scan) and stale ones (matched earlier, no longer matching).
- **Exported columns:** the same set as R1 below, plus `liveNow` and `lastUpdated`. The full row also carries `news`, `catalyst` (with tier, score, evidence and others), `autoBias`, `seenAt`, `shortlistedBy` and `_scoreDetails`, which the card reads directly.

R1 — the day's photo

The only thing the model learns from, and what qp backtests over.

- **When:** the tool's capture minute (one minute after one of its discovery scans, so the photo is fresh).
- **What:** `captureR1` takes **every R0 row of today**, live or stale, and stores each row's **complete JSON** in `r1_frozen.data`. A re-capture on the same day replaces it.
- **Exported columns** (`/api/warehouse/R1/<date>`, CSV export, R2 backups):

Group	Columns	Source
Identity	`ticker`, `date`, `capturedAt`	—
State at the photo	`inShortlist`, `_score`, `bias`	R0
Price	`price`, `prevClose`, `open`, `change`, `gapPct`, `vwap`	TradingView, derived
Moving averages	`sma5`, `ema9`, `ema13`, `ema20`, `ema50`	TradingView
Volatility, volume	`rvol`, `atr`, `adrPct`, `dayHigh`, `dayLow`	TradingView, derived
Ranges	`weekHigh/Low/RangePos`, `monthHigh/Low/RangePos`, `quarterHigh/Low/RangePos`, `yearHigh/Low/RangePos`, `allTimeHigh`	TradingView, derived
Size	`mcap`, `floatShares`	TradingView
Short interest	`shortFloat`, `daysToCover`, `shortBasis`, `shortAsOf`	FINRA / Yahoo / Nasdaq
Pre-market	`pmHigh`, `pmLow`, `pmRange`, `pmAdrRatio`	TradingView, derived
Classification	`sector`, `industry`, `screenerKeys`, `themes`	TradingView, desk
Market context	`regime`, `regimeLabel`, `longTerm`, `midTerm`, `shortTerm`, `broadResolved`, `secBias`, `secScore`, `secHot`	desk (sideD)
Relations (27)	`vsEma9 … vsPmLow`, `distEma9 … distPmLow`, `maStack`, `maStackScore`, `monthQuarter`, `pmAdrBand`, `dayRangePos`, `aboveAllMas`, `belowAllMas`	derived
CANSLIM (31)	`csEpsChg`, `csSalesChg`, `csQuarter`, `csAccelerating`, `csBeats`, `csGrowth3yr`, `csStability`, `csRoe`, `csBaseWeeks`, `csBaseDepth`, `csPivot`, `csPctToPivot`, `csBaseScore`, `csHandle`, `csUdRatio`, `csAdLetter`, `csRsLineTell`, `csGroup`, `csGroupLevel`, `csGroupRank`, `csGroupOf`, `csRsInGroup`, `csGroupMembers`, `csFunds`, `csFundsChg`, `csFundsDir`, `csMarket`, `csDdChecked`, `csDdHeld`, `csDdAvgRel`, `csDdVerdict`	qp
Catalyst	`catalyst` (label only)	news
Cross-tool	`shortlistedElsewhere` (`yes` / `no`)	unified shortlist

Group	Columns	Source
Timing	`foundMinsFromOpen` (earliest screener match, minutes from 09:30; negative = pre-market)	R0 `seenAt`

- **In the stored JSON but not in the exported columns:**
  - `shortSrc`, `shortTrend`;
  - `shortable`, `easyToBorrow`, `regSho`, `regShoDay`;
  - `move2hPct`, `trend6mPct`;
  - the full `catalyst` object, `news`, `seenAt`, `autoBias`, `_changeFromOpen`.

  They are frozen, so the history exists, and a column can be added to the export without losing any past day.

R2 — market snapshots

The market around the photo: eight snapshots between 09:26 and 10:01. Each stores the whole market snapshot (MARKET-TAB.md) as JSON. **Skipped when no scan has built a snapshot yet.**

Group	Columns
When	`date`, `slot` (ET `HH:MM`), `capturedAt`
SPY	`spyClose`, `spyChange`, `spyWeekChg`, `spySma5`, `spySma20`, `spySma50`, `spySma200`, `spyBbUpper`, `spyBbLower`
QQQ	`qqqClose`, `qqqChange`, `qqqWeekChg`, `qqqSma5`, `qqqSma20`, `qqqSma50`, `qqqSma200`
IWM, DIA, VIX	`iwmClose`, `iwmChange`, `iwmWeekChg`, `diaClose`, `diaChange`, `vixClose`, `vixChange`, `vixWeekChg`
Desk read	`regime`, `regimeLabel`, `shortBias`, `shortScore`, `midStage`, `longBias`, `sectorBullish`, `sectorBearish` (counts), `breakouts` (up to 5 symbols up > 3 % on > 2× volume)
O'Neil (read at export time)	`oneilStatus`, `oneilAsOf`, `oneilDistributionDays`, `oneilSessionsSinceFtd`, `oneilFtdDate`
Per sector (15 ×)	`sec_<Sector>_bias`, `sec_<Sector>_score`, `sec_<Sector>_change`, `sec_<Sector>_hot`

The O'Neil columns are read from the current `data/oneil-market.json` when the register is **exported**, not when the snapshot was taken, so an old R2 day shows today's O'Neil status. See *D-D3* in KNOWN-ISSUES.md.

R3A / R3B — what happened after the photo

At 16:05 (`captureR3`), for every ticker in today's R1:

1. **Bars:** 1-minute bars for the day from **Yahoo**, then **Alpaca** for any ticker Yahoo returned empty. Daily bars from Yahoo, else Alpaca. Batches of 100.
2. **ATR14:** from daily bars strictly before today (no look-ahead).
3. **Entry A** = the **open** of the bar at the tool's entry A minute; **Entry B** the same at entry B. No bar at that minute = no row for that ticker (logged as `missing HH:MM bar`).
4. From the entry bar to the close: `hh` = highest high, `ll` = lowest low.
5. `upR = (hh − entry) / ATR14`, `downR = (entry − ll) / ATR14`, both positive.
6. **Opening range** 09:30–09:35 high and low.

Register	Columns
R3A	`date`, `ticker`, `entryPriceA`, `hhA`, `llA`, `atr14`, `upR_A`, `downR_A`, `capturedAt`
R3B	`date`, `ticker`, `entryPriceB`, `hhB`, `llB`, `atr14`, `upR_B`, `downR_B`, `capturedAt`

The opening range is stored (`or_high`, `or_low` in both tables) but not exported — see *D-D4*. A **good move** is `upR ≥ 1.3`.

R4A / R4B — the training rows

R1 joined to R3A (or R3B) on ticker. Read live from the two tables whenever asked; at 16:05 the rows **with an outcome** are also copied into `r4a_train` / `r4b_train`, which is what the model trains on (16:20) and what Analysis exports and accepts as uploads.

Group	Columns
From R1	`date`, `ticker`, `_score`, `inShortlist`, `price936` (R1's `price`; the name is historical, see P-1), `prevClose`, `open`, `change`, `gapPct`, `vwap`, `sma5`, `ema9/13/20/50`, `rvol`, `atr`, `adrPct`, `dayHigh`, `dayLow`, `monthHigh`, `monthLow`, `monthRangePos`, `mcap`, `floatShares`, `shortFloat`, `pmHigh`, `pmLow`, `pmRange`, `pmAdrRatio`, `sector`, `industry`, `screenerKeys`, the market-context columns, `themes`, `bias`, the 27 relation columns, `catalyst`, `shortlistedElsewhere`, `foundMinsFromOpen`
From R3	`entryPrice{A,B}`, `hh{A,B}`, `ll{A,B}`, `atr14`, `upR_{A,B}`, `downR_{A,B}`, `capturedAt`

Narrower than R1. R4 does not carry the 31 CANSLIM columns, the week, quarter and year ranges, `allTimeHigh`, `daysToCover`, `shortBasis` or `shortAsOf`. All of them are in R1 for the same

day. See *D-D2*.

Shortlist

One row per date in `shortlist`, holding a list of items:

Column	Meaning
`date`, `ticker`, `tvSymbol`	—
`addedAt`	when it was starred
`method`	`manual` (the card's ☆) or `auto` (the auto-rule)
`price`, `change`, `sector`, `score`	the card's values at that moment
`exported`, `exportedAt`	copied to TradingView from the Shortlist tab

- **Auto-rule:** one minute before the R1 photo, the top `shortlistTopN` (default 5) cards with a score of at least `shortlistMinScore` (default 70) are added once per day. With the scorer off on every tool, no card has a score, so the auto-rule adds nothing. See *D-D1*.
- **Unified list:** every change is published to `data/shortlist-all.json`; other tools show `★ also T2` on the same ticker. `inShortlist` stays this tool's own decision.

Reading the registers

Where	How
A tool's **Warehouse** tab	pick a date and a register; CSV or JSON export
HTTP	`GET :<tool port>/api/warehouse/available-dates?register=R1`, `GET …/api/warehouse/<REG>/<date>` (login required from outside the box)
qp	reads them over localhost for charts, R1 sheets and backtests
Cloudflare R2	nightly CSV per tool × register × day under `analysis/` (the backup is documented with the system-wide pages)
Archived tools	the archive process serves the same `/api/warehouse` GETs, read-only